

6G Weekly Survey — Sep 07 - Sep 13, 2026

15 items across 4 topics, ranked by importance.

6G Physical Layer

Delay-Doppler Sensing Performance Analysis for MIMO-OFDM ISAC Systems

Peishi Li, Rang Liu, Qian Liu, Ming Li · arXiv eess.SP · 2026-09-01

★★★★■ Addresses a critical knowledge gap in MIMO-OFDM ISAC with analytical insights directly relevant to 6G design, from a credible research group with novel treatment of multi-antenna effects on sensing dynamic range. This paper analyzes delay-Doppler sensing performance in MIMO-OFDM ISAC systems where multiple beamformed data streams jointly illuminate sensing targets. The authors derive second-order moment expressions for range-Doppler maps under matched filtering (MF) and reciprocal filtering (RF), revealing that multi-stream superposition creates an additional RDM floor under MF even with constant-modulus signaling, while RF removes the data-induced floor but amplifies noise according to reciprocal beamforming power. The analysis shows that user-target angular geometry determines the MF/RF tradeoff, with MF more robust under weak illumination and RF achieving higher dynamic range when noise amplification is mild.

Contribution: First rigorous characterization of how spatial beamforming and multi-stream transmission in MIMO-OFDM affect delay-Doppler sensing performance, revealing geometry-dependent MF/RF tradeoffs absent in single-antenna ISAC.

Relevance to your work: Directly relevant to your ISAC research, providing fundamental understanding of how spatial beamforming and multi-stream MIMO-OFDM transmission affects sensing performance—key for designing AI-native ISAC receivers and optimizing joint communication-sensing waveforms.

Wireless Foundation Models: State-of-the-Art and Open Challenges

Alonso M. Pacheco Huachaca, Juan J. Rodriguez Rodriguez, Ahmed Aboufotouh, Nelson L. S. da Fonseca, et al. · arXiv eess.SP · 2026-09-04

★★★★■ This survey provides essential structure to an emerging research area critical for AI-native 6G design, identifying concrete gaps that must be addressed before WFMs can influence commercial systems. This survey systematically analyzes wireless foundation models (WFMs) for physical-layer tasks, organizing the literature into five application families: signal recognition/demodulation, channel representation learning, RF sensing/localization, beam management, and spectrum sensing. The authors examine pretraining strategies, backbone architectures, and downstream adaptation approaches across different distribution shifts (in-distribution, partial-shift, out-of-distribution). The analysis reveals that while current WFMs show promise for learning reusable wireless representations, evaluation inconsistencies and architectural fragmentation make it difficult to identify which design choices drive transferability.

Contribution: First comprehensive taxonomy and critical assessment of the wireless foundation model landscape, highlighting gaps in evaluation rigor and the lack of standardized protocols for assessing true generalization across physical-layer tasks.

Relevance to your work: Directly relevant as it synthesizes work on learned physical layers, neural receivers, channel representation learning, and beam management—core components of AI-native air interface design you are pursuing for beyond-5G systems.

Ultra-High Resolution Method for Multipath Within a Co-Delay-Doppler Bin in DFT-P-OCDM

Mingxuan Han, Weile Zhang, Feifei Gao · arXiv cs.IT · 2026-09-04

★★★★■ Strong technical contribution to ISAC using OCDM waveforms with rigorous analysis (CRLB derivation) and demonstrated performance gains; OCDM variants are serious 6G waveform candidates where resolving fine multipath structure is critical for both sensing and channel estimation.

This paper addresses multipath resolution in DFT-preprocessed orthogonal chirp division multiplexing (DFT-P-OCDM) when multiple physical paths fall into the same delay-Doppler bin, complicating joint communication and sensing. The authors derive input-output relationships in the DFT-preprocessed Fresnel domain and propose a two-stage ultra-high resolution (TSUR) framework: stage one estimates delay from pilot phase progression, stage two estimates Doppler and path count from leakage samples. Simulations show TSUR resolves co-bin paths with lower fractional Doppler error than sequential extraction and off-grid methods, while analysis of Cramér-Rao lower bounds reveals pilot configuration trade-offs between sensing resolution and communication recovery.

Contribution: The novel DPF domain formulation enables ultra-high resolution separation of multipath components within the same delay-Doppler bin for OCDM-based joint communication and sensing, overcoming a fundamental limitation when radar and communication share the same waveform.

Relevance to your work: Directly relevant to your ISAC and 6G waveform research; the super-resolution technique could enhance both sensing accuracy and channel estimation for AI-native receivers operating with chirp-based waveforms in doubly-selective environments.

Energy-Efficient Waveform Design for ISAC Systems: An Ambiguity-Domain QoS Perspective

Ngoc-Son Duong, Trung-Hieu Nguyen, Quang-Truong Can · arXiv eess.SP · 2026-09-04

★★★★ Solid incremental work addressing practical ISAC waveform design with a sensible ambiguity-domain metric, though the approach is algorithmic optimization rather than fundamentally new signal-processing insight.

This paper formulates ISAC waveform design as a transmit-power minimization problem under novel ambiguity-function SINR (AF-SINR) constraints that quantify sensing quality in local delay-Doppler regions, alongside per-user communication-SINR and PAPR constraints. The nonconvex problem (fractional objectives, quartic ambiguity terms, waveform-dependent PAPR) is solved via a fractional-programming successive-convex-approximation algorithm. Simulations demonstrate the method achieves energy efficiency while satisfying communication QoS, localized ambiguity suppression, and practical RF constraints.

Contribution: Introduces AF-SINR as a localized, noise-aware sensing quality metric in the ambiguity domain and develops a tractable algorithm for joint communication-sensing-PAPR-constrained waveform optimization.

Relevance to your work: Directly relevant to your ISAC research: the AF-SINR metric provides a practical handle on sensing-communication tradeoffs, and PAPR awareness is essential for deployable AI-native air interfaces where hardware constraints matter.

Deep Reinforcement Learning for Optimization of STAR-RIS Phase and Energy Splitting Coefficients in OTFS-NOMA Framework

Rais. J. Gachaba, Manobendu Sarker, Anirban Bhowal · arXiv cs.IT · 2026-09-03

★★★★ Solid work combining RIS, OTFS, and DRL for mobility, but limited to simulation with fixed power allocation and no comparison against recent learned-PHY or model-based fast adaptation methods.

The paper addresses real-time reconfiguration of STAR-RIS phase shifts and energy splitting in a downlink OTFS-NOMA system under high Doppler conditions. A deep reinforcement learning agent (Beta-Space Soft Actor-Critic) learns to map channel observations directly to RIS configurations via single forward pass, avoiding per-coherence-interval optimization. Results show ~10% sum-rate degradation across 128-fold speed variation and gains over multiple baselines as transmit power and RIS elements scale.

Contribution: First DRL-based approach for joint STAR-RIS configuration in OTFS-NOMA that handles high-mobility channels without per-interval re-optimization, demonstrating robustness across large user velocity ranges.

Relevance to your work: Directly relevant: combines OTFS waveform optimization with AI/ML for physical layer in scenarios where classical optimization is intractable due to channel dynamics.

Behavior--Realization Separation for Constrained Physical Human--Robot Interaction

Yongyan Cao · arXiv eess.SY · 2026-09-01

★☆☆☆☆ This work focuses on robotics control architecture with no connection to wireless communications, physical-layer signal processing, or radio-frequency system design.

This paper addresses physical human-robot interaction by decoupling desired-behavior specification from constrained realization through a two-layer architecture: a behavior layer defines desired contact-port accelerations, and a realization layer converts these into feasible robot commands via receding-horizon quadratic programming. The approach allows switching between impedance and admittance control without system reconstruction while enforcing constraints across the planning horizon. Experiments on a 7-DOF Franka robot demonstrate sub-millimeter boundary tracking under external forces versus multi-centimeter overshoot from naive clipping, and superior torque-constraint satisfaction compared to single-step enforcement.

Contribution: The separation of behavior and realization layers enables constraint-aware planning that maintains model feasibility across prediction horizons while allowing dynamic behavior switching without re-architecting the control system.

Relevance to your work: tangential

6G Demos & Trials

On the Impact of Site-Specific Training for a Real-World 5G NR System

Reinhard Wiesmayr, Nuri Berke Baytekin, Chris Dick, Christoph Studer · arXiv cs.IT · 2026-09-03

★★★★■ Strong empirical validation from credible group with real hardware, publicly available datasets, and temporal robustness evidence — directly relevant to practical neural receiver deployment in 6G but not yet a fundamental algorithmic breakthrough.

The paper evaluates site-specific finetuning of three neural receiver architectures (fully trainable, model-driven, hybrid model-based) using real 5G NR testbed measurements with dual-layer uplink transmission at ETH Zurich. Results show finetuning substantially improves neural receivers, enables joint training across transmission configurations, and remains effective across campaigns separated by six months. Site-specific LMMSE channel estimation combined with iterative detection achieves the best error rates.

Contribution: First real-world demonstration that site-specific finetuning works for dual-layer 5G NR with standard-compliant signaling, persists across long time separations, and can jointly optimize for multiple transmission modes — with public datasets and code release.

Relevance to your work: Directly relevant: demonstrates practical feasibility of learned PHY components in real multi-layer systems with standard signaling, provides open datasets for benchmarking AI-native air interface designs, and validates temporal stability critical for deployment.

Bayesian thermal digital twin for a space habitat subjected to an impact event

Sreehari Manikkan, Seungho Rhee, Herta Montoya, Davide Ziviani, et al. · arXiv eess.SP · 2026-09-04

★★☆☆■ Competent digital twin implementation for space habitats, but the thermal modeling and Bayesian methods are standard techniques applied to a narrow vertical application outside mainstream 6G research priorities.

This paper develops a Bayesian digital twin for detecting and characterizing micro-meteorite impacts on space habitats by monitoring thermal anomalies in environmental control systems. The approach uses a thermal resistance-capacitance network model with physics-based activation functions for automated model selection, performs offline calibration on physical parameters, then continuously infers impact location, timing, and severity through online Bayesian estimation. Validation on a cyber-physical testbed demonstrates successful anomaly detection, uncertainty-quantified parameter inference, and temperature forecasting for autonomous habitat operation.

Contribution: Novel application of Bayesian digital twin methodology to space habitat thermal management, using continuous inference on impact-sensitive cyber parameters to enable autonomous detection and characterization of structural damage events.

Relevance to your work: Tangential—while digital twins are mentioned in 6G vertical use cases, this work addresses aerospace ECLSS rather than wireless system design, channel modeling, or AI-native air interfaces.

Tuning Collective Patterns to Alleviate Congestion in Shared AI Clusters

Eashan Gupta, Yongzhou Chen, Apoorve Mohan, Pavlos Maniotis, et al. · arXiv cs.NI · 2026-09-03

★☆☆☆☆ Competent networking optimization for AI training infrastructure, but narrowly focused on data center operations rather than wireless PHY or air interface design relevant to 6G systems.

This paper addresses network congestion in shared GPU clusters for distributed AI training by dynamically tuning communication collective patterns (e.g., AllReduce trees) at the application layer. REACT detects congestion using flow statistics and reconfigures which nodes aggregate data while preserving semantic correctness, requiring no network infrastructure changes. Prototyped as a NCCL shim layer, it achieves 13-38% bandwidth improvement in real testbeds and up to 75% in simulations under congestion.

Contribution: Application-layer adaptive tuning of collective communication patterns that can be unilaterally deployed by individual users in shared clouds without requiring global workload coordination or network infrastructure support.

Relevance to your work: tangential

A single-precision floating-point systolic Givens-QRD Triangular Solver for MVDR Beamforming

Athi Ram R S, Alwin A, S. G. Sreejeesh, J. U. Kidav · arXiv eess.SP · 2026-09-02

★☆☆☆☆ Competent hardware implementation study but narrowly scoped to ultrasound medical imaging rather than wireless communications; MVDR is well-established and the contribution is primarily an engineering optimization rather than algorithmic innovation.

This paper implements a systolic Givens-QR decomposition pipeline on FPGA for computing MVDR beamforming weights in ultrasound imaging, using single-precision floating-point arithmetic on a 32-element array. The Zynq UltraScale+ FPGA achieves 31,123 weight vectors/second at 2.45W, demonstrating 2.3x better power-normalized throughput than a 24-core Xeon processor (programmable-logic only), though the CPU retains 15x raw throughput advantage. Numerical validation against MATLAB shows RMSE of 5.10×10^{-4} with 100% pass rate.

Contribution: First detailed FPGA power-performance characterization of systolic Givens-QR for MVDR beamforming with rigorous numerical validation, demonstrating practical energy-efficient acceleration for real-time adaptive array processing.

Relevance to your work: Tangential — while MVDR beamforming is fundamental to massive MIMO and the systolic QR architecture could inform FPGA accelerators for CSI-based precoding or hybrid beamforming, the ultrasound application domain and focus on hardware implementation details rather than wireless channel characteristics limits direct applicability to AI-native air interfaces or 6G PHY research.

SecDT: A Profile-Based Security Layer for TRDP Communications

Erlantz Alonso, Igor Lopez, Jasone Astorga · arXiv cs.NI · 2026-09-02

★☆☆☆☆ This work addresses train control system security (domain-specific industrial protocol) rather than wireless communication systems, making it off-topic for Beyond-5G or 6G physical layer research.

This paper addresses the lack of cryptographic protection in the Train Real-time Data Protocol (TRDP) used in rolling stock by proposing SecDT, a lightweight security layer that supports multiple security profiles from authentication to AEAD encryption. The system uses an on-board key management system for profile negotiation and key distribution, with profiles negotiated per Communication ID to ensure all train subsystems can process secured multicast messages. Prototype implementation using mbedTLS and Arm PSA

demonstrates acceptable overhead for real-time train control systems.

Contribution: The novel contribution is a practical cryptographic agility framework for multicast TRDP communications in safety-critical train systems, balancing security with real-time constraints through dynamic profile negotiation and hardware-backed key management.

Relevance to your work: Tangential - while this addresses industrial IoT security, it focuses on wired train subsystem protocols rather than air interface design, AI-native physical layers, or wireless communication challenges central to 6G research.

AI-native air interface

Closing the Semantic-Edge Gap: Tiny Language Models for 6G Wireless Intelligence

Srikanth Kamath, Arnav Mathur, Joslyn Sajan George, Rahul Jashvantbhai Pandya · arXiv eess.SP · 2026-09-03

★★★★■ Strong synthesis work from emerging topic (semantic communication) with practical edge-deployment focus and credible compression results, though primarily taxonomic rather than algorithmic breakthrough.

This survey connects TinyML compression techniques (quantization, pruning, distillation, LoRA, NAS, hybrids) to semantic communication architectures for 6G, arguing that tiny language models bridge the gap between LLM-level semantic encoding and edge device constraints. A two-axis taxonomy maps six compression families onto five semantic communication architectures, supported by a meta-analysis of size-versus-fidelity trade-offs. Example results include 99.98% model compression for neural MAC protocols and 65% energy reduction using knowledge-graph-assisted transmission.

Contribution: First systematic mapping of TinyML compression methods to semantic communication architectures for 6G edge devices, with a quantitative Pareto-frontier analysis and identification of nine open challenges including two novel to the literature.

Relevance to your work: Directly relevant to AI-native air interfaces and task-oriented communication; the compression-architecture mapping and federated semantic learning connect to your distributed PHY learning and edge intelligence interests.

ISAC (sensing + communication)

AFDM-Enabled ISAC in Dynamic Environments: Fundamentals, Technologies and Opportunities

Linchu Chen, Zhendong Li, Zhou Su, Lin Chen, et al. · arXiv eess.SP · 2026-09-04

★★★★■ AFDM is gaining traction as a 6G candidate waveform, and this thorough treatment from a credible research group provides essential foundation for practitioners working on joint radar-communication systems in mobility-critical scenarios.

This article presents a comprehensive overview of affine frequency-division multiplexing (AFDM) as a waveform for integrated sensing and communication (ISAC) in high-mobility scenarios. AFDM's ability to decouple delay and Doppler in the doubly-dispersive channel makes it advantageous over OFDM for dynamic environments with severe Doppler effects. The work covers fundamental principles, application scenarios, key enabling technologies, and future research directions for AFDM-based ISAC systems.

Contribution: First comprehensive tutorial-style treatment positioning AFDM as a competitive 6G waveform specifically for ISAC under high mobility, systematically covering fundamentals through enabling technologies.

Relevance to your work: Directly relevant: AFDM is among your core 6G waveform interests, and ISAC/JCAS is a primary research area; this provides architectural insight into how novel waveforms enable dual-function systems under Doppler stress.

Feasible but Not Safe: Constraint Violations and Report-Channel Attacks in Learned Cell-Free ISAC Association

★★★★■ Identifies a critical gap between ML training metrics and real-world system requirements for ISAC, providing concrete mitigation strategies (feasibility projection, consistency checks) that will inform deployment of learned schedulers in next-generation distributed systems.

This work examines a learned GNN scheduler (ASSENT) for cell-free ISAC systems and reveals that high prediction accuracy does not guarantee satisfaction of hard constraints (e.g., resource limits, QoS requirements). The authors demonstrate that many GNN outputs violate feasibility constraints and propose greedy projection to restore feasibility. They also expose vulnerability to false data injection attacks where a malicious AP reporting fake statistics can drastically increase infeasible solutions, though attacks affecting objectives can be mitigated while constraint-affecting attacks require consistency checks.

Contribution: First systematic demonstration that learned ISAC schedulers can produce high-accuracy but infeasible solutions, and that they are vulnerable to report-channel attacks that exploit the gap between soft training penalties and hard operational constraints.

Relevance to your work: Directly addresses trustworthiness and operational reliability of AI-native ISAC systems, a key concern for your integrated sensing and communication research as learned schedulers move from simulation to deployment in cell-free architectures.

Task-oriented Framework for Communication-Efficient Federated Learning: From Isolated Optimization to Holistic Synergy

Fuqiang Pan, Yan Liu, Erwu Liu, Arumugam Nallanathan · arXiv eess.SP · 2026-09-04

★★★★■ Solid incremental work connecting federated learning optimization to emerging physical-layer paradigms like ISAC, though the ISAC integration appears limited to resource allocation rather than fundamental PHY-layer redesign.

This paper addresses communication bottlenecks in federated learning by proposing a unified framework integrating model compression (quantization, pruning, low-rank approximation), client selection, and resource allocation, including ISAC and over-the-air computation paradigms. The authors demonstrate a task-oriented design philosophy through autonomous driving case studies showing improved object detection accuracy and reduced training time under constrained communication budgets. The framework couples cross-layer optimization with emerging physical-layer techniques to enable practical FL deployment.

Contribution: The novel contribution is the holistic integration of ISAC and over-the-air computation into federated learning optimization, moving beyond isolated compression or scheduling techniques to joint task-oriented design.

Relevance to your work: Demonstrates practical application of ISAC for resource allocation in distributed ML systems, complementing your work on integrated sensing and communication by showing how ISAC can enable communication-efficient edge intelligence beyond traditional link-budget analysis.